

Problem #1: Stellar timescales (Based on Pols notes 2.5)

By the end of this problem, you should have a solid understanding of the different timescales of how stars respond due to changes of their hydrostatic or thermal equilibrium as well as due to nuclear burning in their centers.

Problem #1a: The nuclear timescale (τ_{nuc})

(i) Calculate the total mass of hydrogen available for fusion over the lifetime of the Sun, if 70% of its mass was hydrogen when the Sun was formed, and only 13% of all hydrogen is in the layers where the temperature is high enough for fusion.

(ii) Derive an expression for the nuclear timescale in solar units, i.e. expressed in terms of $(R/R_{\odot})$, $(M/M_{\odot})$, and $(L/L_{\odot})$.

(iii) Use the mass-radius and mass-luminosity relations for main-sequence stars to express the nuclear timescale of main-sequence stars as a function of the mass of the star only.

(iv) Describe the meaning of the nuclear timescale in your own words.

Problem #1b: The thermal timescale (τ_{therm})

(i) Derive an expression for the thermal timescale in solar units, i.e. expressed in terms of $(R/R_{\odot})$, $(M/M_{\odot})$, and $(L/L_{\odot})$.

(ii) Use the mass-radius and mass-luminosity relations for main-sequence stars to express the thermal timescale of main-sequence stars as a function of the mass of the star only.

(iii) Describe the meaning of the thermal timescale in your own words.

Problem #1c: The dynamical timescale (τ_{dyn})

(i) Derive an expression for the dynamical timescale in solar units, i.e. expressed in terms of $(R/R_{\odot})$, $(M/M_{\odot})$, and $(L/L_{\odot})$.

(ii) Use the mass-radius and mass-luminosity relations for main-sequence stars to express the dynamical timescale of main-sequence stars as a function of the mass of the star only.

(iii) Describe the meaning of the dynamical timescale in your own words.

Problem #1d: Comparing the timescales

(i) Summarize your results for the questions above by computing the nuclear, thermal and dynamical timescales for a 1, 10 and 25 $M_{\odot}$ main-sequence star.

(ii) When the Sun becomes a red giant, its radius will increase to 200 $R_{\odot}$ and its luminosity to 3000 $L_{\odot}$. Estimate τ_{dyn} and τ_{therm} for such a red giant.

(iii) How large would such a RG have to become for $\tau_{\text{dyn}} > \tau_{\text{therm}}$? Assume both R and L increase at constant effective temperature.

Problem #2: Accessing scientific manuscripts as part of astronomy research

The astrophysics data system ([ads](#)) hosted by the Smithsonian Astrophysical Observatory is the central hub of all science manuscripts in astronomy. As part of Project 1, you'll likely need to do some searching to find science papers that are relevant to your topic. This problem is designed to help you get acquainted with ads and to begin your search through the astronomical literature!

Problem #2a: The ads search form

The ads search form contains several examples for searching the literature, including a full text search. The full text search can be very powerful but only if you wield it smartly. Decide on a search term that pertains to your project, report it as an answer, then also report the number of manuscripts that are returned in your search. How useful is this set of literature in bulk?

Problem #2b: Narrowing down a search

Often times, the literature is *too much* to sort through. In this case, you need to use more specific searches to narrow down the results. Based on the available search terms, choose two additional ways to narrow down your search. Report those below and also report the number of manuscripts returned by the search with each additional narrowed down search. Describe in words why you chose the search terms you chose.

Problem #2c: Selecting papers to read

Even with a more manageable set of papers, it can be difficult to choose which papers to start reading. One option is to read the abstracts of the papers returned starting from the most recent until you find a paper that seems relevant. Alternatively, you can start from the oldest paper and read abstracts until you find a paper that seems relevant. Try out both of the options and pick one paper from each option. Which was the easiest option for selection? Why? (Note there isn't a right answer but you should be able to articulate which one you prefer since it will help you in future literature searches!)